MINI PROJECT – S3 2025 Batch

TESTING DOCUMENTATION

SYSTEM TESTING

INTRODUCTION

This section provides an overview of the entire test document. This document describes both the test plan and the test procedure for the **TaskMatrix** web-based multi-company project and task management system with AI risk prediction.

- Unit Testing
- Integration Testing
- System Testing

TEST PLAN

The Test Plan is a detailed document that describes the test strategy, objectives, schedule, estimation, deliverables, and resources required for testing TaskMatrix. The Test Plan helps us determine the effort needed to validate the quality of the application under test. The test plan serves as a blueprint to conduct software testing activities across all application modules, user role boundaries (Global Admin, Company Admin, Project Manager, Developer, Tester), and automated AI risk scoring algorithms.

TEST CASES

The procedure for unit testing, integration testing, and system testing for each software component in TaskMatrix is presented below.

Test Case Field Descriptions:

* **Test Case ID:** Unique identifier for each test case using standard naming conventions (e.g. 'TC_MOD_01', 'TC_AUTH_01').
* **Test Objective:** Brief description of the testing goal.
* **Precondition:** Prerequisites required prior to executing the test case.
* **Steps/Cases:** Ordered sequence of test execution actions.
* **Test Data:** Input values provided during execution.
* **Expected Result:** Output, response, or state change expected after execution.
* **Post-condition:** State of the application/database following test completion.

Test Case ID	Test Objective	Precondition	Steps/Cases	Test Data	Expected Result	Post Condition
TC_MOD_01	Verify Model __str__ Methods	Database active entities created	1. Instantiate model objects. 2. Evaluate str() on each.	GlobalAdmin, Company, CompanyAdmin, Employee objects	Returns formatted label (e.g., 'PM John (Project Manager)')	Model string output verified
TC_MOD_02	Verify Employee Role Properties	Employee records with PM, Dev, Tester roles	1. Query is_project_manager, is_developer, is_tester properties.	Employee instances for PM, Dev, Tester	Returns True for assigned role, False for other roles	Role helper properties validated
TC_MOD_03	Verify Password Hashing Security	Employee record instantiated	1. Call set_password(). 2. Check raw stored password. 3. Test check_password().	Password: 'pass12345', Wrong: 'wrongpass'	Password hashed with PBKDF2; check_password returns True for valid	Plaintext passwords never stored
TC_MOD_04	Verify Module Overdue Calculation	Module records with past and future deadlines	1. Query is_overdue on past deadline in-progress module. 2. Check completed module.	Deadline: today - 1 day, Status: 'In Progress' vs 'Completed'	is_overdue returns True for uncompleted past deadline, False for completed	Overdue logic accurately evaluated
TC_AUTH_01	Root Path Redirection to Login	Unauthenticated browser session	1. Send GET request to root URL '/'	URL: '/'	HTTP 302 redirect to '/login/'	User routed to unified login page
TC_AUTH_02	Company Registration Validations	Registration form accessible	1. Submit mismatched passwords. 2. Submit password < 8 chars. 3. Submit valid form.	Company: 'ValidCo', Email: 'valid@co.com', Pass: 'validpassword'	Errors shown for invalid input; valid registration creates Pending company & redirects	Company record created with status 'Pending'
TC_AUTH_03	Unified Role Login Authentication	Approved companies & active employees in DB	1. POST login for Global Admin. 2. POST login for Company Admin. 3. POST login for Employee.	Credentials for GlobalAdmin, CompanyAdmin, Developer	Session created; user redirected to role-specific dashboard	User authenticated with active role session
TC_AUTH_04	Pending Company Admin Login Block	Company status set to 'Pending'	1. Submit POST login with credentials of pending company admin.	Username: 'padmin', Password: 'pass12345'	Login rejected with message 'company registration is still pending approval'	Unapproved company admin blocked
TC_AUTH_05	User Logout & Session Termination	User currently logged in	1. Send GET request to '/logout/'	Active user session	Redirect to '/login/'; session data completely flushed	Session terminated safely
TC_GADM_01	Global Admin Dashboard Metrics	Logged in as Global Admin	1. Send GET request to '/global-admin/dashboard/'	Global Admin session	HTTP 200 OK; metrics displayed for Total/Pending Companies, Active Employees, Projects	System summary correctly rendered
TC_GADM_02	Pending Company Listing Review	Global Admin logged in; pending company exists	1. Send GET request to '/global-admin/companies/'	Company 'Pending Inc'	HTTP 200 OK; pending company listed with action buttons	Pending requests available for review
TC_GADM_03	Company Approval Workflow	Company status is 'Pending'	1. POST to '/global-admin/companies//approve/'	company_id of Pending company	Company status updated to 'Approved'; decided_at set; redirects to list	Company Admin unlocked for login

Test Case ID	Test Objective	Precondition	Steps/Cases	Test Data	Expected Result	Post Condition
TC_GADM_04	Company Rejection Workflow	Company status is 'Pending'	1. POST to '/global-admin/companies//reject/'	company_id of Pending company	Company status updated to 'Rejected'; login blocked; redirects to list	Registration request rejected
TC_FLOW_01	End-to-End Task & Bug Lifecycle	Approved Company with PM, Dev, Tester	1. Admin assigns project. 2. PM creates module. 3. Dev updates status. 4. Tester reports Bug. 5. Dev fixes Bug. 6. Tester approves.	Project 'Nexus Portal', Module 'Auth', Bug 'Crash'	Module marked 'Completed', project progress set to 100%, Notifications sent	Full task & bug lifecycle verified
TC_RISK_01	AI Risk Prediction - Low Risk	Project created on-track with >50% progress	1. Call calculate_project_risk(project).	Project deadline +20 days, progress 50%	risk_level = 'Low', risk_score = 0, empty risk factors list	Project evaluated low risk
TC_RISK_02	AI Risk Prediction - High Risk	Project with overdue modules & open Critical bugs	1. Call calculate_project_risk(project).	Past deadline, 2 overdue modules, 3 Critical/High bugs	risk_level = 'High', risk_score >= 4, risk factors list populated	Project flagged High Risk on PM dashboard
TC_SEC_01	Role-Based Access Control (RBAC)	Logged in as Developer	1. Dev requests '/company/dashboard/'. 2. Dev requests '/pm/dashboard/'.	Developer session 'dev1'	HTTP 403 Forbidden or redirect to unauthorized error	Unauthorized role access blocked
TC_SEC_02	Cross-Tenant Data Privacy Isolation	Two separate companies (Company 1 & Company 2)	1. Admin from Company 1 requests edit on Company 2 employee.	Company Admin 1 session, Employee 2 ID	HTTP 404 Not Found (tenant scoped database lookup)	Cross-tenant privacy strictly enforced
TC_SEC_03	Employee Deactivation Security	Active employee record	1. Admin deactivates employee. 2. Attempt login as deactivated employee.	Employee username 'dev1'	Status set to 'Inactive'; login attempt rejected with account deactivated error	Account access revoked immediately
TC_SEC_04	Notification Inbox & Mark Read	Unread notification in user inbox	1. User views '/notifications/'. 2. POST to mark read.	Notification ID	Unread counter decremented; status updated to 'Read'	Notification state updated

TEST EXECUTION RESULTS & CONCLUSION

All 20 test cases were executed automatically using Django's test runner framework (`manage.py test`). Every test case passed with zero errors and zero failures (100% pass rate). All critical multi-company boundaries, role authentication, task transitions, bug fixing cycles, and AI risk prediction algorithms operate in strict accordance with the system specification.